A major concern in current UAS regulation is the fragmentation of global legal frameworks. Different countries adopt varying rules for drone registration, permissible altitude, flight range, and operational zones, creating inconsistencies that hinder cross-border operations and global standardization. For instance, while some nations permit beyond visual line of sight (BVLOS) missions under regulated supervision, others restrict drone operations strictly to visual line-of-sight for safety reasons. The absence of harmonized international standards complicates commercial logistics, international research, and cooperative defense operations, where drones must transition across multiple jurisdictions.

Another critical issue lies in airspace integration and traffic management. As the number of drones increases, the risk of airspace congestion and mid-air collisions also rises, especially in urban areas and near airports. To mitigate this, regulatory agencies such as the Federal Aviation Administration (FAA), European Union Aviation Safety Agency (EASA), and the Directorate General of Civil Aviation (DGCA) in India are developing frameworks for unmanned traffic management (UTM) systems. These systems aim to manage drone identification, routing, altitude separation, and communication in real time, ensuring safe and organized coexistence between manned and unmanned aerial vehicles.

Regulatory concerns extend beyond safety to encompass privacy, security, and liability. The use of drones for surveillance or data collection has raised public concerns about unauthorized monitoring and data misuse. Governments and international organizations are now focusing on establishing clear data protection guidelines, geo-fencing protocols, and no-fly zones around sensitive areas to safeguard privacy and national security. Furthermore, determining legal accountability in cases of drone-related